SidewalkPilot Steering Model Report

Dataset-specific current-label offline evaluation

Generated 2026-07-14 19:03:32. Series 1/2 checkpoints are evaluated on their original 2,224-image corrected real field dataset. Series 3/4 checkpoints are evaluated on the same 6,952-image frozen temporal validation subset from the shared 81,237-frame dataset. Rankings and metric gradients are separated by evaluation dataset; Series 1/2 values must not be compared directly with Series 3/4 values. Series 4 uses paired experiment names: p/r for PC, f/g for CF, and a/c for PCF (final/best respectively).

Series 1/2 class-balanced leader: 2.3b, with Bal9 79.2%, turn exact 78.8%, and turn +/-1 98.4%. Lowest Series 1/2 MAE: 2.4b at 3.265 deg.

Series 3/4 class-balanced leader: 4.0p, with Bal9 34.5%, turn exact 32.1%, and turn +/-1 65.9%. Lowest Series 3/4 MAE: 4.0c at 11.321 deg.

Series 1 uses raw BGR and output scale 86. Series 2 uses output scale 85; v2.0/v2.0b use legacy HSV/CLAHE preprocessing, while v2.1 and newer use raw BGR. Series 1/2 produce one continuous steering value. Series 3.0 and 3.0b produce two continuous values (steering and throttle). Series 3.1 and newer produce 19 raw values: 9 steering-class logits, 9 within-class offsets, and throttle. Series 4 removes throttle and emits one or four 18-value hybrid steering horizons; PC/PCF also consume three previous targets. Every architecture is decoded to current steering degrees before the common metrics are calculated.

Series 1/2 Evaluation Sources

Source	Count	Purpose
D0328_17	315	First dataset relabel, March 28
D0329_15	413	First dataset relabel, March 29
D0425_14	65	Street test
D0426_18	53	Curves and shadows
D0427_18	72	Curved curb
D0429_19	53	Driveway and shadow
D0502_12	154	Shadow fix set
D0502_19	156	Hard turns, curb hugging, smoothness
D0503_17	159	Harsh sidewalk
D0506_20	24	8pm sidewalk failure set
D0510_18	167	v2.3 field-failure run 1: turns, road-right driving, driveways
D0510_19	8	v2.3 field-failure run 2 (short)
D0510_20	585	v2.3 field-failure run 3: turns, road-right driving, driveways

Series 3/4 Evaluation Sources

Source	Count	Purpose
D0702_16	4230	Series 3 July 2 base set: two manual field runs, 50,684 images
D0707_16	652	Bright-sun HARD/diagonal shadows across the sidewalk (v3.2 shadow-robustness batch)
D0712_16	2070	Series 3 manual field collection, July 12 run 2 (103-frame bad segment removed)

Series 1/2 Class-Balanced Model Ranking

Ranked by macro-average exact recall across the 9 steering buckets, then turn within-one-bucket recall, then MAE. Each metric column has its own red-yellow-green scale: green means better within this table. Signed error is greenest nearest zero.

Rank	Model	Checkpoint filename	Prep	Bal9	Turn exact	Turn +/-1	ST exact	MAE	Med	Signed
1	2.3b	SidewalkPilot-v2.3b.pth	raw BGR	79.2%	78.8%	98.4%	62.8%	3.41	2.05	0.41
2	2.4b	SidewalkPilot-v2.4b.pth	raw BGR	78.2%	75.9%	99.5%	55.0%	3.27	2.39	-0.04
3	2.3	SidewalkPilot-v2.3.pth	raw BGR	77.8%	76.6%	98.5%	63.0%	3.30	1.93	0.01
4	2.4	SidewalkPilot-v2.4.pth	raw BGR	77.5%	74.9%	99.5%	55.5%	3.28	2.46	-0.23
5	2.2	SidewalkPilot-v2.2.pth	raw BGR	64.6%	62.4%	92.4%	42.3%	6.68	4.26	-0.22
6	2.2b	SidewalkPilot-v2.2b.pth	raw BGR	64.3%	62.1%	92.1%	42.3%	6.68	4.23	-0.46
7	2.1	SidewalkPilot-v2.1.pth	raw BGR	54.2%	51.0%	79.4%	39.4%	10.80	5.83	-1.15
8	2.0b	SidewalkPilot-v2.0b.pth	HSV/CLAHE -> BGR	54.2%	50.0%	78.8%	38.9%	11.13	5.07	-2.62
9	2.1b	SidewalkPilot-v2.1b.pth	raw BGR	54.1%	50.9%	79.6%	38.2%	10.75	5.80	-1.12
10	2.0	SidewalkPilot-v2.0.pth	HSV/CLAHE -> BGR	53.8%	49.5%	78.9%	38.5%	11.12	5.09	-2.63
11	1.9b	SidewalkPilot-v1.9b.pth	raw BGR	43.2%	41.8%	75.3%	37.3%	12.94	6.60	-2.06
12	1.9	SidewalkPilot-v1.9.pth	raw BGR	43.1%	41.3%	75.1%	36.9%	12.98	6.63	-2.06
13	1.8b	SidewalkPilot-v1.8b.pth	raw BGR	38.7%	35.8%	69.6%	36.0%	15.41	9.36	-0.70
14	1.8	SidewalkPilot-v1.8.pth	raw BGR	38.6%	35.8%	69.5%	36.1%	15.47	9.46	-0.65
15	1.7b	SidewalkPilot-v1.7b.pth	raw BGR	35.4%	35.1%	69.9%	37.5%	15.54	9.53	-0.87
16	1.7	SidewalkPilot-v1.7.pth	raw BGR	35.0%	34.9%	69.2%	37.3%	15.82	9.66	-0.48
17	1.5	SidewalkPilot-v1.5.pth	raw BGR	30.8%	30.7%	63.9%	33.3%	16.82	10.59	-0.89
18	1.5b	SidewalkPilot-v1.5b.pth	raw BGR	30.8%	30.7%	63.9%	33.3%	16.82	10.59	-0.89
19	1.4	SidewalkPilot-v1.4.pth	raw BGR	30.5%	29.4%	63.3%	36.6%	16.81	10.19	-3.99
20	1.6b	SidewalkPilot-v1.6b.pth	raw BGR	30.0%	30.5%	64.9%	30.5%	17.41	11.04	-0.01

Rank	Model	Checkpoint filename	Prep	Bal9	Turn exact	Turn +/-1	ST exact	MAE	Med	Signed
21	1.6	SidewalkPilot-v1.6.pth	raw BGR	29.6%	29.6%	64.7%	29.9%	17.53	11.21	0.21
22	1.4b	SidewalkPilot-v1.4b.pth	raw BGR	29.0%	28.5%	62.2%	40.0%	17.04	10.07	-4.15
23	1.2	SidewalkPilot-v1.2.pth	raw BGR	27.8%	27.4%	61.5%	32.9%	18.64	11.79	-3.55
24	1.2b	SidewalkPilot-v1.2b.pth	raw BGR	27.4%	26.6%	62.4%	31.6%	18.63	11.85	-3.73
25	1.3	SidewalkPilot-v1.3.pth	raw BGR	27.0%	26.2%	62.5%	30.4%	18.55	11.81	-3.43
26	1.3b	SidewalkPilot-v1.3b.pth	raw BGR	26.9%	27.0%	62.5%	32.2%	18.48	11.24	-3.65
27	1.1	SidewalkPilot-v1.1.pth	raw BGR	24.7%	24.1%	56.1%	26.1%	20.85	13.42	-5.16
28	1.1b	SidewalkPilot-v1.1b.pth	raw BGR	24.7%	24.1%	56.1%	26.1%	20.85	13.42	-5.16
29	1.0b	SidewalkPilot-v1.0b.pth	raw BGR	17.6%	17.1%	47.1%	20.6%	25.07	17.45	-7.08
30	1.0	SidewalkPilot-v1.0.pth	raw BGR	15.3%	15.5%	45.1%	18.9%	25.55	18.46	-7.45

Series 3/4 Class-Balanced Model Ranking

Ranked by macro-average exact recall across the 9 steering buckets, then turn within-one-bucket recall, then MAE. Each metric column has its own red-yellow-green scale: green means better within this table. Signed error is greenest nearest zero.

Rank	Model	Checkpoint filename	Prep	Bal9	Turn exact	Turn +/-1	ST exact	MAE	Med	Signed
1	4.0p	SidewalkPilot-v4.0p.onnx	raw BGR 320x180	34.5%	32.1%	65.9%	67.7%	12.40	2.97	0.12
2	4.0a	SidewalkPilot-v4.0a.onnx	raw BGR 320x180	33.5%	30.9%	65.3%	68.1%	12.38	3.11	0.29
3	4.0r	SidewalkPilot-v4.0r.onnx	raw BGR 320x180	32.9%	27.4%	62.6%	77.6%	11.64	1.85	-1.14
4	4.0c	SidewalkPilot-v4.0c.onnx	raw BGR 320x180	32.0%	29.4%	62.9%	75.5%	11.32	1.83	-0.98
5	3.1	SidewalkPilot-v3.1.onnx	raw BGR 320x180	28.8%	27.2%	55.5%	55.7%	22.23	9.44	-3.11
6	3.1b	SidewalkPilot-v3.1b.onnx	raw BGR 320x180	28.2%	25.2%	53.8%	57.1%	20.59	9.33	-4.51
7	3.2	SidewalkPilot-v3.2.onnx	raw BGR 320x180	26.1%	24.3%	55.1%	51.0%	17.46	9.73	-1.45
8	4.0f	SidewalkPilot-v4.0f.onnx	raw BGR 320x180	25.4%	23.5%	56.4%	62.8%	15.62	6.72	1.06
9	3.4	SidewalkPilot-v3.4.onnx	raw BGR 320x180	24.2%	22.6%	56.2%	64.2%	15.08	6.07	0.42
10	3.3	SidewalkPilot-v3.3.onnx	raw BGR 320x180	23.8%	18.5%	47.7%	65.3%	15.17	6.13	-5.41
11	3.4b	SidewalkPilot-v3.4b.onnx	raw BGR 320x180	22.4%	19.1%	51.2%	72.7%	13.99	2.48	-1.57
12	3.2b	SidewalkPilot-v3.2b.onnx	raw BGR 320x180	21.0%	15.4%	43.9%	67.6%	14.64	5.13	-4.54
13	4.0g	SidewalkPilot-v4.0g.onnx	raw BGR 320x180	20.4%	17.1%	46.4%	76.0%	14.12	2.11	-1.86
14	3.3b	SidewalkPilot-v3.3b.onnx	raw BGR 320x180	19.0%	8.6%	36.9%	79.2%	16.08	1.90	-8.19
15	3.0	SidewalkPilot-v3.0.onnx	raw BGR 320x180	16.7%	16.5%	45.8%	23.5%	18.19	13.12	-2.28
16	3.0b	SidewalkPilot-v3.0b.onnx	raw BGR 320x180	16.2%	15.6%	44.6%	26.3%	17.57	12.63	-2.59

Chronological Model Growth

S1/2 and S3/4 rows use different evaluation sets. Compare chronology and metrics only within the same eval-set group.

Model	Checkpoint filename	Series	Eval set	Prep	Scale	MAE	Median	Signed	<=2	<=5	<=20	Pred mean
1.0	SidewalkPilot-v1.0.pth	1	S1/2	raw BGR	86	25.55	18.46	-7.45	130	333	1184	88.87
1.0b	SidewalkPilot-v1.0b.pth	1	S1/2	raw BGR	86	25.07	17.45	-7.08	135	351	1227	89.24
1.1	SidewalkPilot-v1.1.pth	1	S1/2	raw BGR	86	20.85	13.42	-5.16	243	510	1416	91.16
1.1b	SidewalkPilot-v1.1b.pth	1	S1/2	raw BGR	86	20.85	13.42	-5.16	243	510	1416	91.16
1.2	SidewalkPilot-v1.2.pth	1	S1/2	raw BGR	86	18.64	11.79	-3.55	267	582	1519	92.77
1.2b	SidewalkPilot-v1.2b.pth	1	S1/2	raw BGR	86	18.63	11.85	-3.73	275	584	1547	92.59
1.3	SidewalkPilot-v1.3.pth	1	S1/2	raw BGR	86	18.55	11.81	-3.43	261	557	1545	92.89
1.3b	SidewalkPilot-v1.3b.pth	1	S1/2	raw BGR	86	18.48	11.24	-3.65	265	578	1553	92.67
1.4	SidewalkPilot-v1.4.pth	1	S1/2	raw BGR	86	16.81	10.19	-3.99	320	676	1620	92.33
1.4b	SidewalkPilot-v1.4b.pth	1	S1/2	raw BGR	86	17.04	10.07	-4.15	336	685	1598	92.17
1.5	SidewalkPilot-v1.5.pth	1	S1/2	raw BGR	86	16.82	10.59	-0.89	375	666	1572	95.43
1.5b	SidewalkPilot-v1.5b.pth	1	S1/2	raw BGR	86	16.82	10.59	-0.89	375	666	1572	95.43
1.6	SidewalkPilot-v1.6.pth	1	S1/2	raw BGR	86	17.53	11.21	0.21	316	630	1556	96.53
1.6b	SidewalkPilot-v1.6b.pth	1	S1/2	raw BGR	86	17.41	11.04	-0.01	318	629	1562	96.31
1.7	SidewalkPilot-v1.7.pth	1	S1/2	raw BGR	86	15.82	9.66	-0.48	415	744	1615	95.84
1.7b	SidewalkPilot-v1.7b.pth	1	S1/2	raw BGR	86	15.54	9.53	-0.87	410	769	1635	95.45
1.8	SidewalkPilot-v1.8.pth	1	S1/2	raw BGR	86	15.47	9.46	-0.65	478	796	1645	95.67
1.8b	SidewalkPilot-v1.8b.pth	1	S1/2	raw BGR	86	15.41	9.36	-0.70	492	794	1649	95.62
1.9	SidewalkPilot-v1.9.pth	1	S1/2	raw BGR	86	12.98	6.63	-2.06	584	966	1728	94.26
1.9b	SidewalkPilot-v1.9b.pth	1	S1/2	raw BGR	86	12.94	6.60	-2.06	597	979	1734	94.26
2.0	SidewalkPilot-v2.0.pth	2	S1/2	HSV/CLAHE -> BGR	85	11.12	5.09	-2.63	609	1068	1839	93.70
2.0b	SidewalkPilot-v2.0b.pth	2	S1/2	HSV/CLAHE -> BGR	85	11.13	5.07	-2.62	601	1064	1838	93.70
2.1	SidewalkPilot-v2.1.pth	2	S1/2	raw BGR	85	10.80	5.83	-1.15	477	982	1867	95.18
2.1b	SidewalkPilot-v2.1b.pth	2	S1/2	raw BGR	85	10.75	5.80	-1.12	480	978	1867	95.20
2.2	SidewalkPilot-v2.2.pth	2	S1/2	raw BGR	85	6.68	4.26	-0.22	594	1238	2094	96.10
2.2b	SidewalkPilot-v2.2b.pth	2	S1/2	raw BGR	85	6.68	4.23	-0.46	613	1255	2094	95.86
2.3	SidewalkPilot-v2.3.pth	2	S1/2	raw BGR	85	3.30	1.93	0.01	1129	1768	2194	96.33

Model	Checkpoint filename	Series	Eval set	Prep	Scale	MAE	Median	Signed	<=2	<=5	<=20	Pred mean
2.3b	SidewalkPilot-v2.3b.pth	2	S1/2	raw BGR	85	3.41	2.05	0.41	1088	1742	2195	96.73
2.4	SidewalkPilot-v2.4.pth	2	S1/2	raw BGR	85	3.28	2.46	-0.23	957	1724	2218	96.09
2.4b	SidewalkPilot-v2.4b.pth	2	S1/2	raw BGR	85	3.27	2.39	-0.04	953	1729	2218	96.28
3.0	SidewalkPilot-v3.0.onnx	3	S3/4	raw BGR 320x180	-	18.19	13.12	-2.28	570	1402	4707	94.00
3.0b	SidewalkPilot-v3.0b.onnx	3	S3/4	raw BGR 320x180	-	17.57	12.63	-2.59	642	1544	4827	93.69
3.1	SidewalkPilot-v3.1.onnx	3	S3/4	raw BGR 320x180	-	22.23	9.44	-3.11	2571	3000	4439	93.16
3.1b	SidewalkPilot-v3.1b.onnx	3	S3/4	raw BGR 320x180	-	20.59	9.33	-4.51	2657	3062	4539	91.77
3.2	SidewalkPilot-v3.2.onnx	3	S3/4	raw BGR 320x180	-	17.46	9.73	-1.45	2414	2751	4686	94.82
3.2b	SidewalkPilot-v3.2b.onnx	3	S3/4	raw BGR 320x180	-	14.64	5.13	-4.54	3112	3461	5184	91.74
3.3	SidewalkPilot-v3.3.onnx	3	S3/4	raw BGR 320x180	-	15.17	6.13	-5.41	2949	3360	4970	90.87
3.3b	SidewalkPilot-v3.3b.onnx	3	S3/4	raw BGR 320x180	-	16.08	1.90	-8.19	3542	3899	5073	88.08
3.4	SidewalkPilot-v3.4.onnx	3	S3/4	raw BGR 320x180	-	15.08	6.07	0.42	2943	3372	4974	96.69
3.4b	SidewalkPilot-v3.4b.onnx	3	S3/4	raw BGR 320x180	-	13.99	2.48	-1.57	3367	3762	5198	94.71
4.0p	SidewalkPilot-v4.0p.onnx	4	S3/4	raw BGR 320x180	-	12.40	2.97	0.12	3268	3717	5452	96.40
4.0r	SidewalkPilot-v4.0r.onnx	4	S3/4	raw BGR 320x180	-	11.64	1.85	-1.14	3598	4110	5526	95.14
4.0f	SidewalkPilot-v4.0f.onnx	4	S3/4	raw BGR 320x180	-	15.62	6.72	1.06	2900	3308	4838	97.33
4.0g	SidewalkPilot-v4.0g.onnx	4	S3/4	raw BGR 320x180	-	14.12	2.11	-1.86	3405	3868	5233	94.41
4.0a	SidewalkPilot-v4.0a.onnx	4	S3/4	raw BGR 320x180	-	12.38	3.11	0.29	3252	3713	5454	96.57
4.0c	SidewalkPilot-v4.0c.onnx	4	S3/4	raw BGR 320x180	-	11.32	1.83	-0.98	3556	4016	5577	95.30

Series 1/2 Field-Case / Subset MAE

Lower is better. All rows in this table use the same evaluation dataset.

Model	D0328_17	D0329_15	D0425_14	D0426_18	D0427_18	D0429_19	D0502_12	D0502_19	D0503_17	D0506_20	D0510_18	D0510_19	D0510_20
2.3b	2.45	2.57	3.41	3.15	3.42	2.52	3.46	3.23	2.50	4.00	4.47	2.79	4.60
2.4b	3.16	2.92	4.11	4.04	4.08	2.51	4.62	3.58	2.78	2.47	3.13	1.83	3.15
2.3	2.39	2.51	3.30	3.04	3.46	2.54	3.23	3.07	2.38	3.71	4.20	2.65	4.50
2.4	3.14	2.96	4.09	3.97	4.12	2.48	4.66	3.60	2.84	2.48	3.21	1.69	3.16
2.2	3.84	3.62	5.42	4.70	5.66	4.39	5.68	4.74	3.53	2.50	14.71	8.43	10.51
2.2b	3.81	3.56	5.46	4.97	5.61	4.38	5.67	4.72	3.50	2.62	14.65	8.25	10.58
2.1	19.70	20.13	3.69	3.26	3.89	3.32	4.14	3.62	2.47	3.60	10.64	6.88	8.76
2.0b	18.09	20.54	2.69	2.02	2.57	1.70	2.32	2.25	1.66	35.80	11.71	7.49	10.54
2.1b	19.65	19.96	3.65	3.19	3.81	3.33	4.10	3.64	2.42	3.63	10.67	6.86	8.73
2.0	17.97	20.51	2.71	2.03	2.57	1.69	2.35	2.26	1.66	36.00	11.70	7.45	10.59
1.9b	17.10	21.79	1.67	1.49	1.73	1.43	1.36	1.78	28.24	35.89	10.51	5.75	10.87
1.9	17.27	21.89	1.66	1.52	1.74	1.41	1.32	1.79	28.23	35.45	10.48	5.70	10.91
1.8b	17.43	20.92	1.52	1.19	1.42	0.97	1.08	22.57	28.02	37.04	12.89	5.38	14.69
1.8	17.40	20.92	1.51	1.18	1.42	0.96	1.08	22.62	28.08	37.19	13.08	5.39	14.87
1.7b	16.52	20.73	1.39	1.03	1.23	1.01	17.43	22.69	27.59	30.87	10.19	5.57	12.68
1.7	16.55	21.03	1.44	1.12	1.23	0.93	17.51	22.84	27.33	30.78	10.21	5.76	13.50
1.5	16.69	20.32	1.24	0.88	0.99	25.74	18.59	21.31	26.42	26.32	10.75	6.48	15.96
1.5b	16.69	20.32	1.24	0.88	0.99	25.74	18.59	21.31	26.42	26.32	10.75	6.48	15.96
1.4	16.34	20.23	0.88	0.56	31.68	25.83	18.73	24.90	26.43	34.05	10.71	6.82	11.14
1.6b	17.01	20.01	1.24	0.90	1.10	26.22	19.02	22.90	27.36	29.09	10.00	5.66	17.51
1.6	16.84	20.09	1.26	0.84	1.10	26.32	18.90	22.95	27.36	29.10	9.99	5.64	17.99
1.4b	16.94	20.62	1.00	0.72	31.91	26.65	19.21	24.99	26.13	34.16	10.42	7.00	11.30
1.2	16.38	19.88	0.87	39.03	31.76	25.39	20.86	23.31	32.51	36.88	14.69	6.91	11.84
1.2b	16.39	19.86	0.93	38.42	32.11	25.46	21.01	23.36	32.64	36.64	14.57	6.80	11.77
1.3	16.46	20.12	0.87	37.56	34.06	25.27	19.51	23.50	34.04	32.99	13.71	6.20	11.46
1.3b	16.63	20.54	1.02	36.75	35.63	24.12	19.66	23.53	34.63	32.20	13.01	6.25	10.81
1.1	17.12	20.33	1.32	41.55	36.60	33.44	21.42	26.71	33.09	29.72	13.52	5.50	17.34

Model	D0328_17	D0329_15	D0425_14	D0426_18	D0427_18	D0429_19	D0502_12	D0502_19	D0503_17	D0506_20	D0510_18	D0510_19	D0510_20
1.1b	17.12	20.33	1.32	41.55	36.60	33.44	21.42	26.71	33.09	29.72	13.52	5.50	17.34
1.0b	17.36	20.30	49.19	45.40	54.06	37.39	29.15	30.30	40.51	40.61	17.89	15.52	18.26
1.0	18.07	21.08	49.02	46.31	53.69	37.44	29.56	30.14	41.72	40.76	17.95	14.47	18.74

Series 3/4 Field-Case / Subset MAE

Lower is better. All rows in this table use the same evaluation dataset.

Model	D0702_16	D0707_16	D0712_16
4.0p	9.69	11.40	18.24
4.0a	9.68	11.13	18.29
4.0r	9.70	9.86	16.15
4.0c	9.48	10.34	15.39
3.1	15.15	27.26	35.12
3.1b	13.25	26.21	33.82
3.2	15.33	20.39	20.89
4.0f	12.08	17.89	22.15
3.4	11.77	16.06	21.55
3.3	11.72	21.05	20.36
3.4b	10.86	19.15	18.74
3.2b	11.83	21.18	18.33
4.0g	10.85	16.88	19.92
3.3b	11.08	18.46	25.53
3.0	16.85	16.60	21.41
3.0b	16.00	16.67	21.08

Prediction Distribution

Model	Pred min	P05	P25	Median	P75	P95	Pred max	Pred mean	Target mean
2.3b	5.00	58.64	83.54	93.98	104.04	164.76	175.00	96.73	96.32
2.4b	5.00	57.77	82.46	93.29	103.57	164.71	175.00	96.28	96.32
2.3	5.00	58.42	83.26	93.64	103.24	165.00	175.00	96.33	96.32
2.4	5.00	57.47	82.19	93.08	103.48	164.71	175.00	96.09	96.32
2.2	5.00	51.21	81.67	93.65	105.09	167.06	175.00	96.10	96.32
2.2b	5.00	51.20	81.50	93.35	104.86	167.64	175.00	95.86	96.32
2.1	5.00	46.92	79.86	93.56	105.96	161.84	175.00	95.18	96.32
2.0b	5.00	41.26	77.43	92.44	106.38	160.18	175.00	93.70	96.32
2.1b	5.00	47.17	79.95	93.49	105.85	161.30	175.00	95.20	96.32
2.0	5.00	41.35	77.37	92.45	106.30	160.31	175.00	93.70	96.32
1.9b	4.00	46.67	78.77	92.67	107.24	157.44	176.00	94.26	96.32
1.9	4.00	46.17	78.77	92.52	107.33	157.92	176.00	94.26	96.32
1.8b	4.00	48.10	80.15	94.64	109.16	154.91	176.00	95.62	96.32
1.8	4.00	48.29	80.12	94.60	109.19	155.61	176.00	95.67	96.32
1.7b	4.00	49.29	81.17	94.08	108.57	153.13	176.00	95.45	96.32
1.7	4.00	49.39	81.31	94.44	109.12	153.53	176.00	95.84	96.32
1.5	4.00	47.33	80.30	93.99	108.59	153.09	176.00	95.43	96.32
1.5b	4.00	47.33	80.30	93.99	108.59	153.09	176.00	95.43	96.32
1.4	4.00	48.35	80.64	92.55	102.58	136.88	176.00	92.33	96.32
1.6b	4.00	48.23	79.75	94.62	109.73	154.52	176.00	96.31	96.32
1.6	4.00	48.57	80.01	94.65	109.88	154.94	176.00	96.53	96.32
1.4b	4.00	48.10	80.18	92.30	101.93	137.30	176.00	92.17	96.32
1.2	4.00	47.80	78.33	93.44	107.11	136.20	176.00	92.77	96.32
1.2b	4.00	47.73	78.53	93.35	106.81	137.56	176.00	92.59	96.32
1.3	4.00	47.72	79.07	93.19	107.25	133.68	176.00	92.89	96.32
1.3b	4.00	47.97	79.35	92.96	106.07	134.30	176.00	92.67	96.32
1.1	4.00	44.02	75.57	90.82	106.24	141.57	176.00	91.16	96.32
1.1b	4.00	44.02	75.57	90.82	106.24	141.57	176.00	91.16	96.32

Model	Pred min	P05	P25	Median	P75	P95	Pred max	Pred mean	Target mean
1.0b	9.01	43.91	72.06	88.65	104.89	136.78	173.93	89.24	96.32
1.0	8.46	42.83	71.11	88.86	105.18	136.92	174.15	88.87	96.32
4.0p	7.25	69.57	88.71	90.02	92.07	152.14	170.05	96.40	96.28
4.0a	5.87	69.28	88.78	90.06	92.45	152.39	171.42	96.57	96.28
4.0r	7.66	79.29	88.75	89.86	91.44	149.89	169.99	95.14	96.28
4.0c	7.46	79.14	88.92	90.00	91.33	149.83	165.81	95.30	96.28
3.1	1.12	17.09	80.83	89.66	99.73	157.98	179.06	93.16	96.28
3.1b	1.19	18.03	80.79	89.84	99.41	156.15	178.85	91.77	96.28
3.2	1.36	67.28	80.43	90.08	99.95	155.52	177.91	94.82	96.28
4.0f	5.34	68.12	88.66	89.97	99.93	152.24	174.96	97.33	96.28
3.4	2.86	68.98	88.69	90.03	92.41	150.96	177.49	96.69	96.28
3.3	2.03	67.61	88.61	90.08	91.64	127.68	173.96	90.87	96.28
3.4b	8.12	68.66	88.85	89.91	91.25	148.82	170.13	94.71	96.28
3.2b	1.36	68.32	88.82	90.11	91.72	113.33	178.38	91.74	96.28
4.0g	9.04	79.43	88.54	89.62	90.99	148.98	171.81	94.41	96.28
3.3b	7.24	24.76	88.81	89.85	91.13	111.71	175.42	88.08	96.28
3.0	20.05	66.32	82.25	93.94	104.79	123.70	166.77	94.00	96.28
3.0b	25.02	67.29	83.31	93.49	104.30	119.92	159.07	93.69	96.28

Prediction Buckets vs Ground

Bucket columns show prediction counts; exact/off-by-one compare prediction bucket against ground bucket.

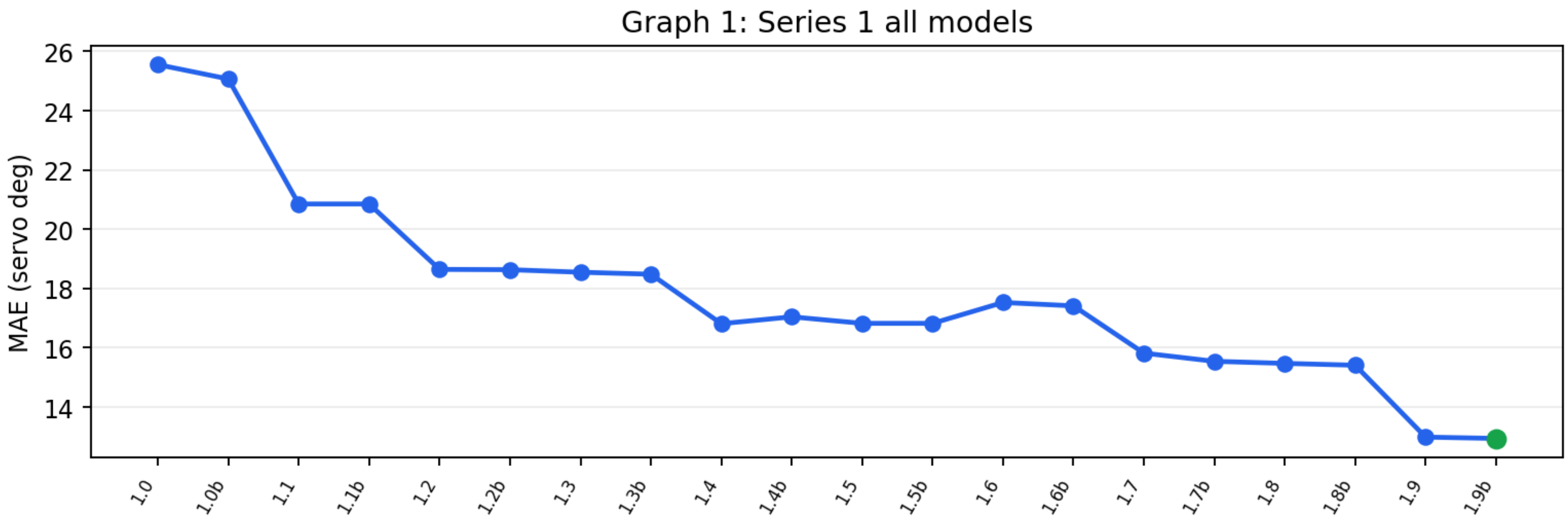
Model	HL 0-45	L 45-75	SL 75-85	ST 85-95	SR 95-105	R 105-135	HR 135-180	Exact	Off by <=1
2.3b	74	173	372	559	531	307	208	75.7%	98.2%
2.4b	76	199	391	543	507	296	212	71.2%	98.5%
2.3	73	181	380	578	514	293	205	74.3%	98.2%
2.4	76	209	389	552	498	292	208	70.8%	98.4%
2.2	94	244	366	488	473	342	217	58.9%	92.3%
2.2b	94	252	371	493	465	330	219	58.6%	92.0%
2.1	97	331	303	456	455	358	224	50.7%	84.8%

Model	HL 0-45	L 45-75	SL 75-85	ST 85-95	SR 95-105	R 105-135	HR 135-180	Exact	Off by <=1
2.0b	123	373	310	432	394	373	219	49.7%	84.7%
2.1b	96	327	309	449	458	361	224	50.2%	84.9%
2.0	123	375	310	430	395	374	217	49.3%	84.8%
1.9b	107	352	320	453	387	400	205	44.0%	81.9%
1.9	106	354	322	451	388	399	204	43.4%	81.8%
1.8b	99	332	278	426	404	481	204	40.0%	78.6%
1.8	99	332	276	424	404	479	210	40.1%	78.6%
1.7b	92	313	280	470	418	456	195	39.3%	79.6%
1.7	93	313	271	463	403	477	204	39.1%	78.9%
1.5	99	315	304	446	393	465	202	35.1%	76.9%
1.5b	99	315	304	446	393	465	202	35.1%	76.9%
1.4	86	308	326	545	482	357	120	34.4%	75.4%
1.6b	92	331	290	420	413	440	238	34.2%	76.8%
1.6	91	339	282	420	405	445	242	33.5%	76.5%
1.4b	94	300	313	589	470	333	125	34.5%	75.0%
1.2	92	382	282	428	424	497	119	32.6%	74.4%
1.2b	93	394	273	433	425	483	123	31.7%	74.3%
1.3	88	365	310	432	398	523	108	30.8%	73.5%
1.3b	85	342	322	455	421	491	108	31.8%	74.0%
1.1	119	425	326	420	344	449	141	27.7%	68.4%
1.1b	119	425	326	420	344	449	141	27.7%	68.4%
1.0b	121	530	308	382	329	432	122	21.1%	58.2%
1.0	134	528	327	345	328	440	122	19.6%	56.2%
4.0p	36	329	1018	3873	178	897	621	58.9%	81.4%
4.0a	28	365	931	3906	251	831	640	59.0%	81.3%
4.0r	75	255	624	4509	77	966	446	64.6%	81.7%
4.0c	38	299	848	4404	110	684	569	63.4%	82.8%
3.1	637	364	762	3291	257	729	912	48.2%	68.3%
3.1b	588	380	793	3398	214	860	719	48.9%	69.1%

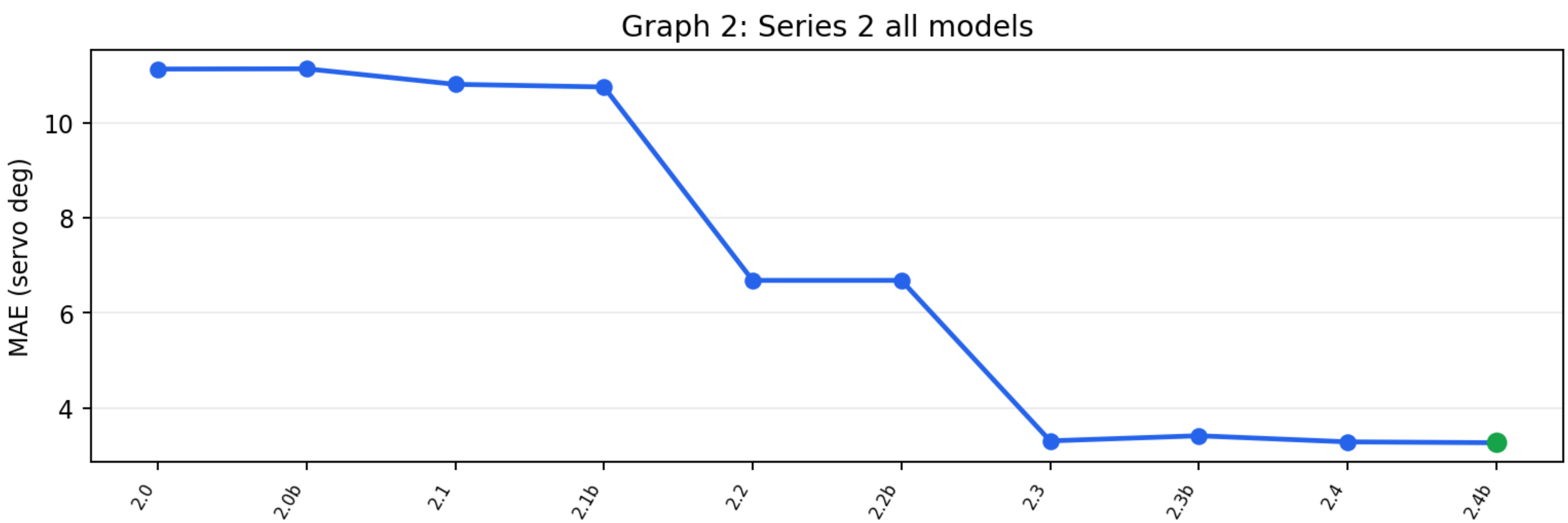
Model	HL 0-45	L 45-75	SL 75-85	ST 85-95	SR 95-105	R 105-135	HR 135-180	Exact	Off by <=1
3.2	178	760	877	3003	758	617	759	44.3%	71.8%
4.0f	79	460	807	3742	188	843	833	52.4%	74.1%
3.4	79	384	923	3858	168	782	758	53.1%	75.5%
3.3	174	694	588	4069	548	536	343	52.2%	73.0%
3.4b	59	376	764	4526	34	668	525	57.6%	77.1%
3.2b	171	270	852	4200	585	580	294	52.7%	77.1%
4.0g	77	166	842	4767	105	347	648	58.4%	77.8%
3.3b	465	205	170	5182	416	311	203	57.8%	73.7%
3.0	30	848	1234	1554	1569	1621	96	23.8%	63.1%
3.0b	16	807	1173	1753	1564	1570	69	25.4%	64.4%

Graphs

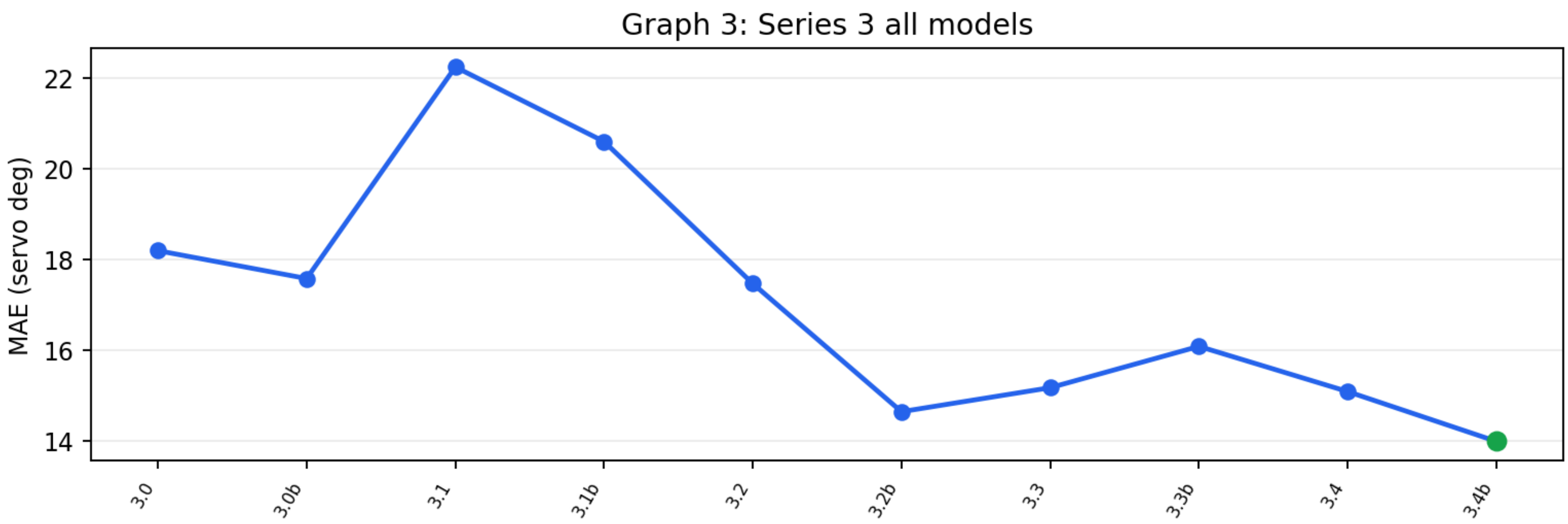
Graph 1: Series 1 all models



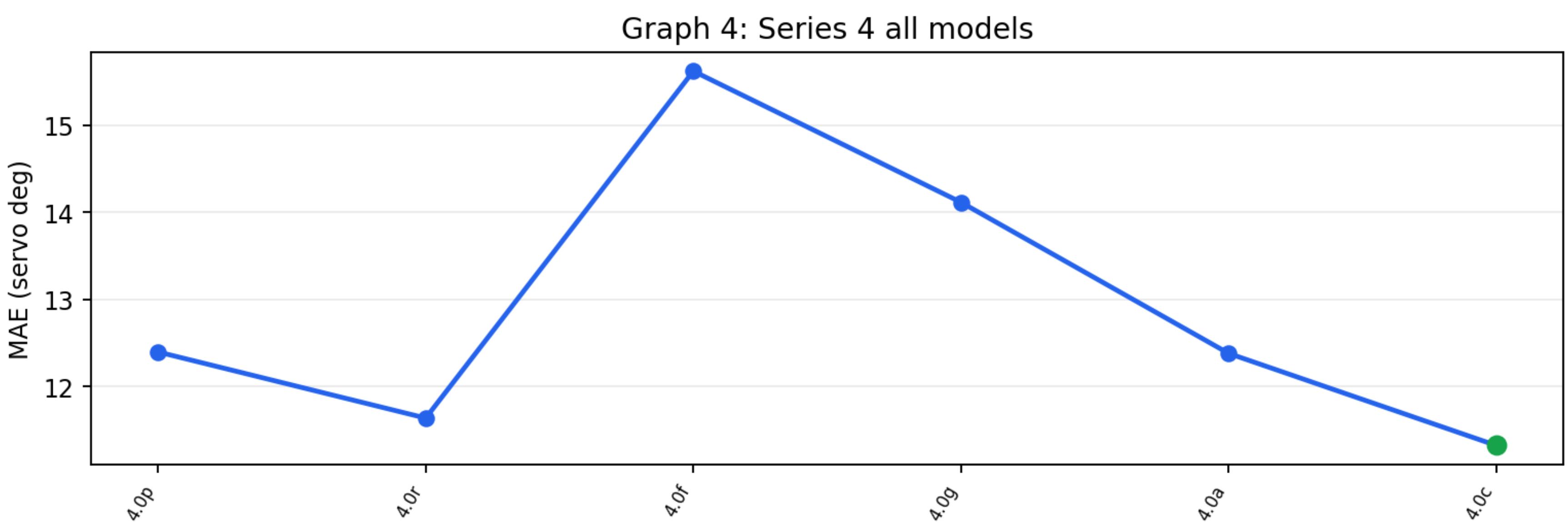
Graph 2: Series 2 all models



Graph 3: Series 3 all models



Graph 4: Series 4 all models



Steering-Bucket Confusion (9-class) - latest Series 3/4 models

Rows = the TRUE steering bucket, columns = the model's PREDICTED bucket. Green diagonal = correct; red off-diagonal = confusion. Shading = share of that true bucket (each row sums to ~100%). Buckets: HL 0-45, L 45-60, L+ 60-75, SL 75-85, ST 85-95, SR 95-105, R 105-120, R+ 120-135, HR 135-180.

v4.0c · exact bucket 61% · within one bucket 80%

T\P	HL	L	L+	SL	ST	SR	R	R+	HR
HL	18	4	6	6	15	1	·	·	·
L	6	3	36	20	23	·	·	·	·
L+	1	·	85	66	70	·	1	·	·
SL	2	·	57	185	112	·	·	·	1
ST	11	6	102	563	3579	51	159	61	209
SR	·	·	·	4	124	15	54	16	26
R	·	·	·	2	133	15	75	39	51
R+	·	·	·	2	105	11	104	63	77
HR	·	·	·	·	243	17	64	48	205

v4.0a · exact bucket 56% · within one bucket 78%

T\P	HL	L	L+	SL	ST	SR	R	R+	HR
HL	17	7	5	10	10	·	·	·	1
L	1	12	32	25	18	·	·	·	·
L+	·	5	92	67	59	·	·	·	·
SL	1	2	61	194	96	3	·	·	·
ST	8	16	132	627	3230	139	228	62	299
SR	·	·	1	4	105	27	53	14	35
R	·	·	·	1	109	31	78	43	53
R+	1	·	·	·	84	28	105	78	66
HR	·	·	·	3	195	23	82	88	186

v4.0g · exact bucket 57% · within one bucket 75%

T\P	HL	L	L+	SL	ST	SR	R	R+	HR
HL	4	4	7	13	20	·	1	·	1
L	4	6	8	32	28	1	·	·	9
L+	5	17	28	85	81	1	·	2	4
SL	13	12	15	142	166	1	·	3	5
ST	46	26	38	539	3604	36	63	91	298
SR	·	1	·	8	169	4	8	12	37
R	·	·	·	8	200	8	19	19	61
R+	1	1	1	2	197	31	30	21	78
HR	4	·	2	13	302	23	42	36	155

v4.0f · exact bucket 50% · within one bucket 70%

T\P	HL	L	L+	SL	ST	SR	R	R+	HR
HL	13	5	11	3	14	·	·	·	4
L	2	5	34	22	17	·	·	1	7
L+	5	6	74	70	60	1	·	1	6
SL	7	5	85	132	116	2	5	·	5
ST	45	12	216	552	2976	86	342	113	399
SR	1	1	1	7	122	12	33	22	40
R	·	·	2	7	133	21	46	31	75
R+	·	·	2	3	108	43	80	33	93
HR	6	·	1	11	196	23	105	31	204

v4.0r · exact bucket 62% · within one bucket 79%

T\P	HL	L	L+	SL	ST	SR	R	R+	HR
HL	19	8	2	7	14	·	·	·	·
L	7	18	17	18	27	1	·	·	·
L+	2	8	68	65	77	1	1	·	1
SL	3	2	41	170	141	·	·	·	·
ST	42	26	65	361	3681	47	260	56	203
SR	·	·	·	3	116	9	81	8	22
R	·	·	·	·	128	8	110	31	38
R+	1	·	·	·	107	3	148	66	37
HR	1	·	·	·	218	8	134	71	145

v4.0p · exact bucket 56% · within one bucket 78%

T\P	HL	L	L+	SL	ST	SR	R	R+	HR
HL	17	8	4	8	13	·	·	·	·
L	2	15	28	26	16	1	·	·	·
L+	·	2	90	73	57	·	1	·	·
SL	2	1	62	203	87	·	·	·	2
ST	14	18	101	698	3209	90	244	64	303
SR	·	·	·	5	106	22	65	15	26
R	·	·	·	4	105	23	93	41	49
R+	1	·	·	1	85	17	106	90	62
HR	·	·	·	·	195	25	94	84	179

v3.4b · exact bucket 56% · within one bucket 74%

T\P	HL	L	L+	SL	ST	SR	R	R+	HR
HL	4	4	20	6	11	·	·	1	4
L	8	1	29	17	26	·	·	·	7
L+	7	2	88	46	77	·	·	·	3
SL	9	6	70	110	156	·	·	·	6
ST	26	4	142	558	3449	14	93	191	264
SR	1	·	·	8	162	2	11	23	32
R	·	·	·	4	186	2	31	48	44
R+	·	·	5	3	181	11	57	63	42
HR	4	1	4	12	278	5	76	74	123

v3.4 · exact bucket 51% · within one bucket 72%

T\P	HL	L	L+	SL	ST	SR	R	R+	HR
HL	11	2	16	8	11	·	1	·	1
L	5	·	29	29	19	·	2	·	4
L+	5	1	76	76	64	·	·	·	1
SL	10	·	73	146	116	3	4	2	3
ST	42	3	176	630	3046	68	272	140	364
SR	1	·	1	10	138	5	26	25	33
R	·	·	3	3	143	22	36	41	67
R+	1	·	3	3	104	39	74	39	99
HR	4	·	1	18	217	31	78	42	186

Notes

- Series 1/2 checkpoints were evaluated on 2,224 original corrected real images. Series 3/4 checkpoints were evaluated on the same 6,952 sequence-valid frozen validation anchors. Cross-group values are not directly comparable.

- Series 2 v2.0/v2.0b used their required HSV/CLAHE preprocessing; all other Series 1/2 models used raw BGR.

- MAE and within-degree counts can reward straight collapse. Use the class-balanced and turn-recall columns for selection, with MAE as supporting evidence.

- Offline MAE does not prove real-world reliability. The car can still fail on lighting, turns, driveways, road-edge ambiguity, speed, and sensor conditions.

- Series 1/2 training was CARLA-assisted, but this report follows the historical evaluator and scores them on the 2,224 corrected real-image set rather than synthetic training frames.

- The Series 3/4 report set is held out by the frozen 100-frame window split and requires three valid previous and three valid future frames without split crossings or timestamp gaps.

- The shared source dataset contains 81,237 curated real field frames across five manual-driving runs from July 2 through July 12, 2026; 6,952 anchors satisfy the common report contract.

- The hold-last baseline repeats the most recent previous target. It is a persistence reference for Series 4 history models, not a deployed controller behavior.

- Series 3 throttle labels are near-constant at full throttle, so this report evaluates steering only; throttle control remains disabled pending varied-throttle data.

- The July 13 field comparison selected v3.4 from the cases presented. Series 4 checkpoints remain experimental and require deployment-compatible integration and field testing.